



AMD
Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by **I2S**

Team Details

- Team name – Escanor
- Team leader name – Sumesh Ranjan Majee
- Problem Statement – Ai in Education and Skilling

Brief about the idea

Personalise learning with explainable, multilingual support that boosts confidence and outcomes.

CourseForge is an Agentic AI Learning Platform that instantly generates structured, university-grade curriculums on any topic. It personalizes the learning journey with multi-lingual audio summaries, interactive code sandboxes, and adaptive quizzes, powered by a resilient multi-model AI architecture

Opportunities

-> How different is it from existing ideas?

1. Vs. ChatGPT: ChatGPT gives unstructured text walls. CourseForge builds a LMS-grade hierarchy (Modules -> Lessons -> Quizzes) with state persistence.
2. Vs. Udemy/Coursera: CourseForge is dynamic. It creates content on-demand for niche topics (e.g., "Rust for React Developers") that don't have pre-recorded courses yet.
3. Vs. Static Tutorials: Includes active recall (quizzes) and hands-on practice (sandboxes), not just passive reading.

-> How will it be able to solve the problem?

1. Personalization: "Pro Mode" uses Chain-of-Thought reasoning to tailor depth to the user's level.
2. Access: Breaks language barriers with "Hinglish" and regional language audio summaries.
3. Engagement: Reduces anxiety via bite-sized lessons and instant feedback loops.

-> USP (Unique Selling Proposition):

1. Self-Healing AI: If the AI generates invalid data, a background agent fixes it automatically (Zero-Crash reliability).
2. Multilingual Voice: "Podcast-style" summaries for learning on the go.
3. Hybrid Architecture: Combines fast caching (Redis) with deep reasoning (Llama 3 70B) for cost-effective scale.

List of features offered by the solution

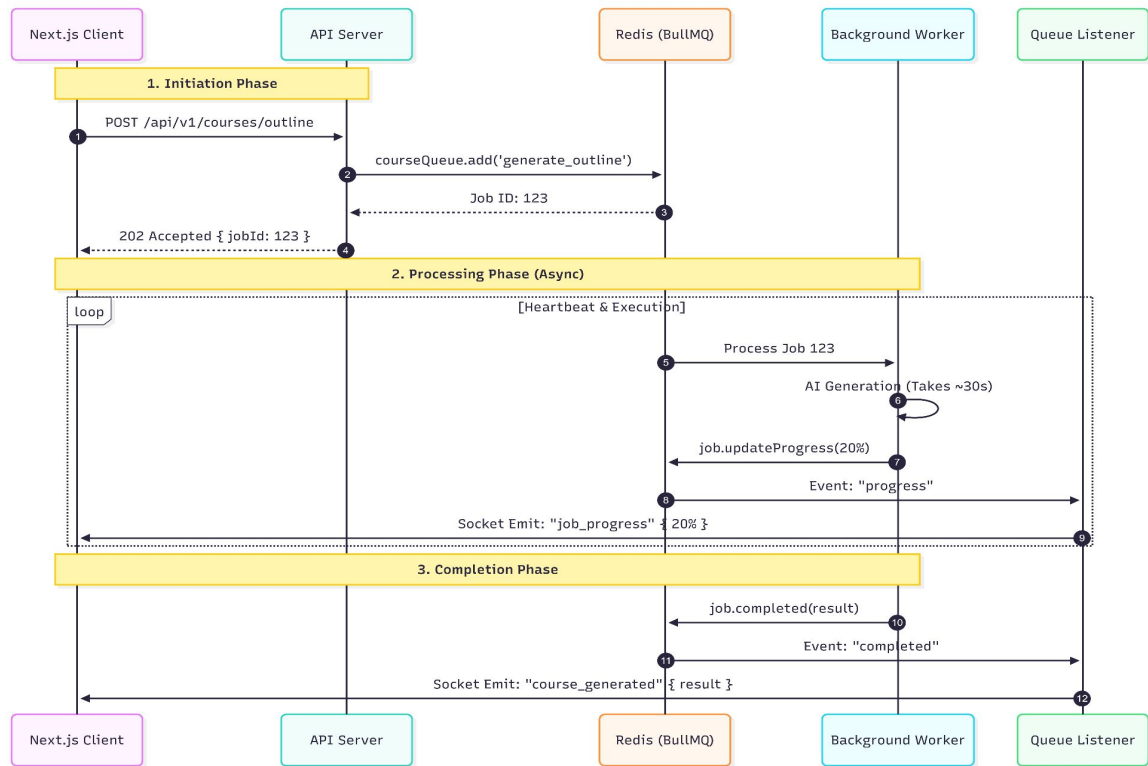
1. Instant Curriculum Architect: Generates structured syllabi (Modules & Chapters) in seconds.
2. Pro Mode (Reasoning Engine): Uses advanced CoT (Chain of Thought) to explain complex topics "First Principles" style.
3. Multilingual Audio Studio: Converts text lessons into natural-sounding audio (English, Hindi, Hinglish, Tamil, etc.).
4. Interactive Code Sandbox: Integrated Python/JS environment where students can run code and get AI-driven debugging help.
5. Visual Enrichment: Auto-curates relevant diagrams and YouTube videos for every lesson.
6. Knowledge Checks: Auto-generated MCQs at the end of modules to ensure mastery.

Process flow diagram or Use-case diagram

Description:

1. User Request: Student asks for "Quantum Computing Course".
2. Orchestration: API offloads task to Redis Queue (Non-blocking).
3. Agentic Workers: Background workers pick up the job, select the best AI model (Llama 3 via AMD ROCm context), and generate content.
4. Real-Time Push: Progress updates sent to User via [Socket.io](https://socket.io).
5. 5. Delivery: Full course rendered in UI + saved to MongoDB.

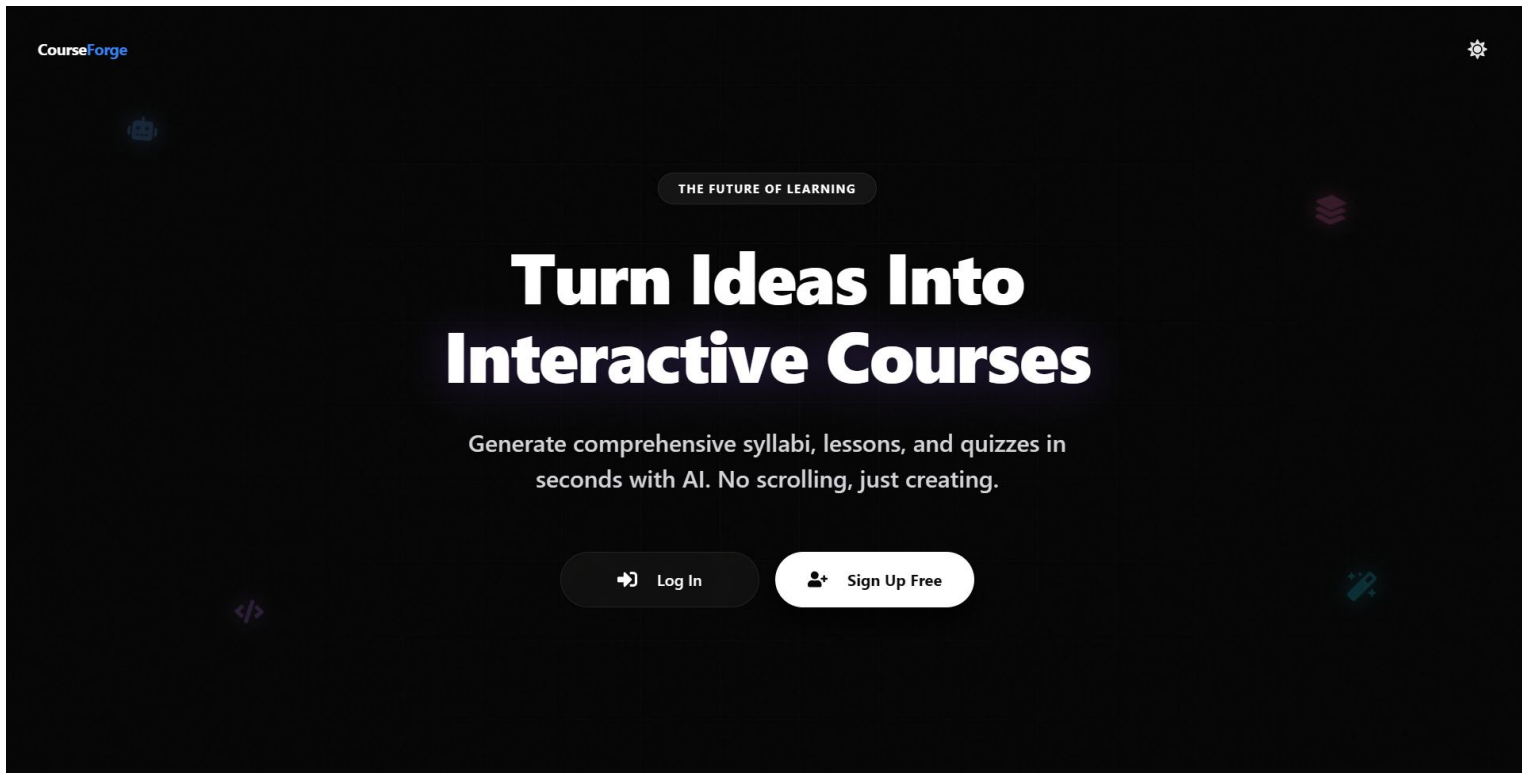
Asynchronous Event Loop



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Clean, distraction-free interface designed for focus. Dark mode enabled for late-night study sessions



Dashboard

The screenshot shows the CourseForge dashboard. At the top, there's a navigation bar with a hamburger menu, the 'CourseForge' logo with a 'PRO' badge, a settings gear, a document icon with '1565' and a plus sign, and a user profile icon with the letter 'T'. On the left is a sidebar with a 'New Chat' button, a 'Dashboard' button (highlighted), and a 'My Library' button. Below these are lists of topics under 'TODAY' and 'YESTERDAY' sections. The main area features a large heading 'What do you want to master today?' with 'master today?' in pink. Below this is a description: 'AI-driven curriculum design. Enter a topic, and we'll engineer a structured learning path tailored to your goals.' A search bar contains the placeholder text 'Describe a complex topic (e.g. 'Advanced System Design')...' with a right arrow button. Below the search bar are a 'Pro Mode' toggle (which is turned on) and a '100 Credits' badge. At the bottom, there are three feature cards: 'Deep Research' (with a globe icon), 'Reasoning Engine' (with a brain icon), and 'Expert Verification' (with a diamond icon). Each card has a brief description of its capabilities.

☰ CourseForge PRO

⚙️ 1565 + T

+ New Chat

🏠 Dashboard

📖 My Library

TODAY

Computer Networks

Mongodb Essentials and Advance...

Database Management Systems

Intermediate Backend...

Advance System Design

Mongodb Essentials and Advance...

Database Management Systems

Database Management Systems

Mongodb Essentials and Advance...

Database Management Systems

YESTERDAY

SQL Query Optimization and...

⚙️ Settings

What do you want to master today?

AI-driven curriculum design. Enter a topic, and we'll engineer a structured learning path tailored to your goals.

Describe a complex topic (e.g. 'Advanced System Design')... ➔

☒ Pro Mode 🔒 100 Credits

Deep Research
Scans live web sources to build the most up-to-date curriculum available.

Reasoning Engine
Uses advanced models (DeepSeek/GPT-4) to solve complex logic and coding tasks.

Expert Verification
Includes interactive clarification steps to tailor the course specifically to you.

My Learning

CourseForge PRO

1565

T

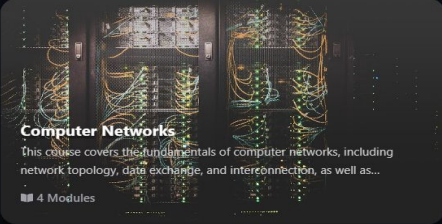
+

🏠

📖

My Library

🔍 Search your library...



Computer Networks

This course covers the fundamentals of computer networks, including network topology, data exchange, and interconnection, as well as...

4 Modules



Mongodb Essentials and Advanced Topics

A comprehensive course that covers MongoDB's core libraries, key concepts, and best practices, along with real-world usage examples...

5 Modules

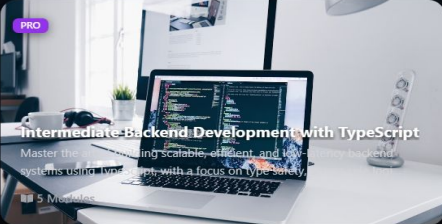


Database Management Systems

Understand the fundamental of designing and implementing database management systems, including key concepts and best practices...

5 Modules

PRO

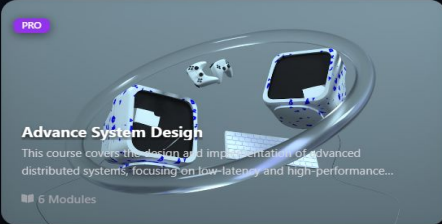


Intermediate Backend Development with TypeScript

Master the art of building scalable, efficient, and robust backend systems using TypeScript, with a focus on type safety and performance.

5 Modules

PRO



Advance System Design

This course covers the design and implementation of advanced distributed systems, focusing on low-latency and high-performance...

6 Modules



Mongodb Essentials and Advanced Topics

A comprehensive course that covers MongoDB's core libraries, key concepts, and best practices, along with real-world usage examples...

5 Modules

PRO



What do you want to know?

...

...

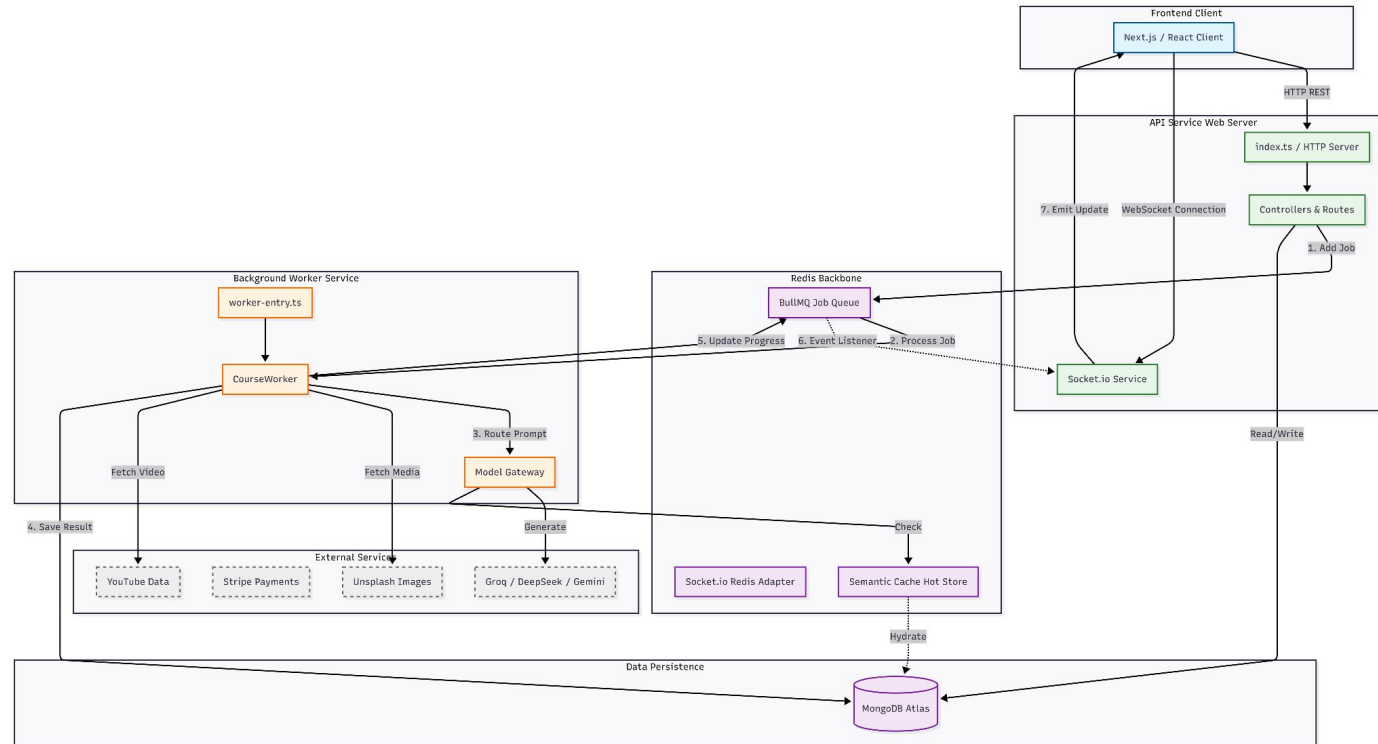
PRO



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Architecture diagram of the proposed solution



Technologies to be used in the solution

1. Core Frameworks: React, Express.js, TypeScript, Chakra Ui
2. AI Models (Hybrid Tiering):
 3.
 - a. Reasoning & Pro Mode: DeepSeek-V3 / GPT-4o (for complex logic & "Pro" generation).
 - b. Fast Inference: Llama 3 (8B & 70B) via Groq (for standard lessons & summaries).
 - c. Repair Agent: Gemini 1.5 Flash (for JSON syntax healing).
3. Infrastructure: Docker, Redis (BullMQ), Socket.io.
4. External Services:
 - a. Stripe: Payments & Subscriptions.
 - b. Cloudinary: Media asset management.
 - c. Tavily: AI Research & Web Search Context.
 - d. Unsplash: Context-aware image curation.
 - e. Piston: Secure code execution sandbox.

Usage of AMD Products/Solutions

1. Inference Optimization with AMD ROCm™:
CourseForge relies heavily on Llama 3. We leverage the AMD ROCm™ open software platform to optimize inference throughput. By running our 'Standard Mode' (Llama 3 8B) on AMD Instinct™ MI300 series accelerators, we can achieve lower latency for real-time course generation compared to standard CUDA paths.
2. Local AI on Ryzen™ AI:
Future Roadmap: We plan to port the 'Student Copilot' (chat assistant) to run locally on Ryzen™ AI PCs (NPU). This ensures student data privacy and offline capability, utilizing the dedicated AI engine in Ryzen 7000/8000 series processors.

Estimated implementation cost

1. Infrastructure: ~\$25/month (Render/VPS for hosting).
2. AI Inference: ~\$0.05 per course generated
3. Storage: Free tier (MongoDB Atlas) to start; scales with user growth.
4. AMD Advantage: Switching to AMD Instances (e.g., Azure NG-series) can reduce inference costs by up to 30% for our Llama 3 workloads.

AI cost in detail

Variable Costs (Per Generated Course)

Based on LangSmith Token Tracing (Avg. 5 Lessons/Course)

Item	Standard Mode (Llama-3/Groq)	Pro Mode (DeepSeek/GPT-4o)
Token Load	~7,200 Tokens	~20,000 Tokens (High Context)
AI Cost	~\$0.002 (Groq/Llama-8B)	~\$0.03 - \$0.12 (DeepSeek/GPT)
Search (Tavily)	\$0 (Skipped)	\$0.005 (Web Research Step)
Images (Unsplash)	Free (API Tier)	Free (API Tier)
Total Cost	< \$0.01 / Course	~\$0.04 - \$0.15 / Course

Prototype Assets

- GitHub Repository: https://github.com/SRMajee/CourseForge_New
- Live Demo: <https://courseforgemajee.vercel.app/>
- Video Demo:
https://drive.google.com/file/d/1D3c1_BuJ4Fu8te0TLTlaRxP2DfEolxoW/view?usp=sharing

Why CourseForge Wins

Education isn't one-size-fits-all. CourseForge uses the power of Generative AI to make high-quality, personalized education accessible to anyone, in their own language, at their own pace. Built for the future, powered by AMD



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Slingshot

HUMAN *IMAGINATION*
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Thank you!
